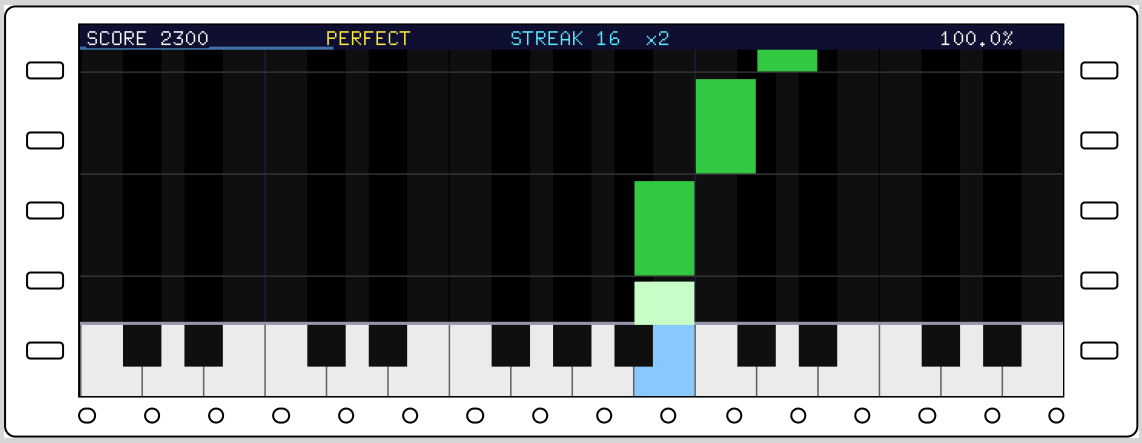
PIANO GAME

Keyfall

for the KN7000

Operating Instructions



Homebrew software for the Technics SX-KN7000; not a Technics or Panasonic publication.
Edition of 30 October 2026, for the program in kn5000_homebrew/demos/kn7000_keyfall (the folder KEYFALL\ on the SD card).
Program and manual: GPLv3 or later. The songs supplied are public-domain pieces arranged for the game. Headings set in Archivo (SIL Open Font License 1.1).

Introduction

Notes fall; you play them

Keyfall shows a keyboard along the bottom of the display, each key right above its own key of the keybed, and a song's notes falling down towards it. Strike each key as its note reaches the keyboard. Every key you strike is judged by its timing: **PERFECT**, **GOOD**, **OK**, or a **WRONG KEY** if no note was due on it; a note nobody strikes is **MISSED**. A score, a streak and an accuracy count up as you go, and a results page with a grade follows the song.

The songs are Standard MIDI Files on the SD card. You choose the part you play: a track of the song, or both hands when it has a track for each. The other parts are the *accompaniment*: Keyfall sends them to the instrument as a sequencer on its **MIDI IN** would, and the instrument plays them with its own sounds.

The keybed keeps its sound. Keyfall only listens to it, so whatever sound the panel has chosen for the keybed is what you play the song with, and the rest of the panel works as always.

■ What goes on the SD card

The folder KEYFALL\, beside the program menu, holds:

file	what it is
KN7APP.BIN	the link between the game and the instrument's firmware
KEYFALL.BIN	the game
MANIFEST.TXT	its name in the program menu: "Keyfall (piano game)", in Games
***.MID**	the songs
SCORES.TXT	the best scores, written by the game
KFLOG.TXT	a log of what the game did, for finding faults

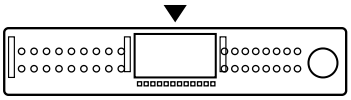
Starting and leaving

1 Press **DISK MENU**, then **STYLE CONVERT**, and **OK** with the loader floppy in the drive. The program menu appears.

2 Open the **Games** folder, choose "Keyfall (piano game)" with the button beside it, and press **RUN**. After a few seconds the song list appears.

3 To leave, hold **EXIT** for a second and a half. The instrument shows its own message and asks "Continue Next?": **NO** returns to the instrument, **YES** starts the program menu again.

The song list





The song list, Ode to Joy selected.

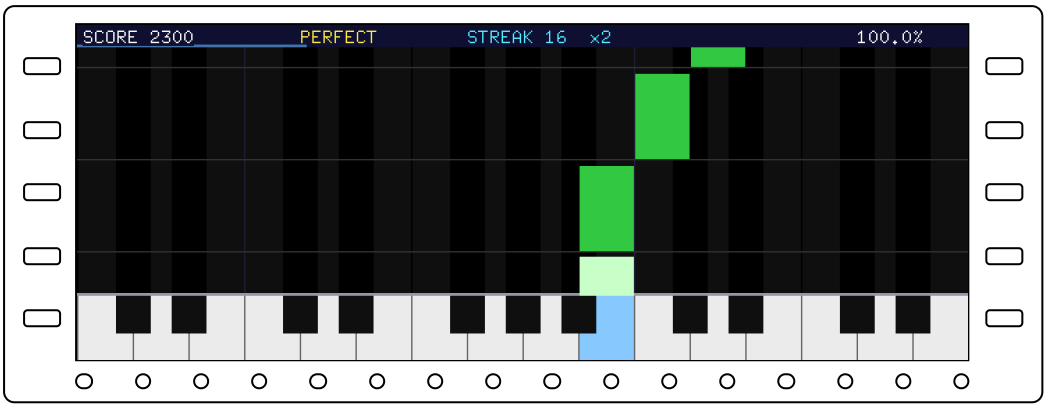
The left column lists the MIDI files in the folder; the right one shows the song selected (its title, length and number of parts) and how it will be played. Turn the **TEMPO/PROGRAM** wheel, or press **LCDR 2** (UP) and **LCDR 3** (DOWN), to choose a song; the soft keys set the rest.

button	label	what it does
LCDL 1	PART	The part you play. Each track of the song in turn, and "Both hands" when one track is named for the right hand and one for the left.
LCDL 2	TEMPO	The song's speed: 100%, 90% … 50%. A slower tempo spreads the notes out; the judging's timing stays the same.
LCDL 3	ACCOMP	The accompaniment (the parts you do not play) on or off.
LCDL 4	MODE	**PLAY**: the game, judged. **WAIT**: the song stops at each note until you strike its key (for learning a piece). **DEMO**: the song plays by itself, your part too.
LCDL 5	SPEED	How fast the notes fall: 70 to 180 lines a second. Faster shows less of what comes, but spaces the notes more.
LCDR 1	PLAY	Starts the song.
LCDR 5	LATENCY	Milliseconds taken off the time of every key before it is judged (-40 to 100). If the results say you are late on average although you play in time, raise it by that much.

THE KEYBED'S LEFT SIDE

Keyfall reads the keys the keybed sends to **MIDI OUT** (the parts RIGHT 1, RIGHT 2 and LEFT). While **AUTO PLAY CHORD** is on, the keys left of the split point are the accompaniment's chord keys and send nothing: parts for the right hand play as they are; for a left-hand part, or both hands, turn **AUTO PLAY CHORD** off first.

Playing a song



A song playing: the notes of the right hand fall in green; the key that has just been struck correctly is lit green on the keyboard.

■ The screen

- **The top line**: the score, the last judgement (PERFECT, GOOD, OK, MISS, WRONG KEY), the streak with its multiplier, and the accuracy so far. Under it, a thin line grows with the song's progress.
- **The lanes**: one for each key shown, the black keys' darker. A note's length shows how long it is held. Its lower edge reaches the keyboard at the moment to strike it. The faint lines across are the beats, the brighter ones the first beats of the bars.
- **The keyboard**: at least two octaves, as many as the part needs. A key turns pale green just before a note is due on it, blue while you hold it, green when you struck a note, red for a wrong key.
- The first notes start falling two and a half seconds after **PLAY**. With two parts, the second (the left hand) falls in blue.

■ Judging

key struck	judgement	points
within 50 ms of the note	PERFECT	100
within 100 ms	GOOD	70
within 150 ms	OK	40
no note due on that key	WRONG KEY	0
a note not struck in time	MISS	0

The points are multiplied by 1 plus a tenth of the streak (the notes struck in a row), at most by 4. A miss or a wrong key ends the streak. The accuracy counts each note as 100, 70, 40 or 0 per cent; wrong keys do not lower it.

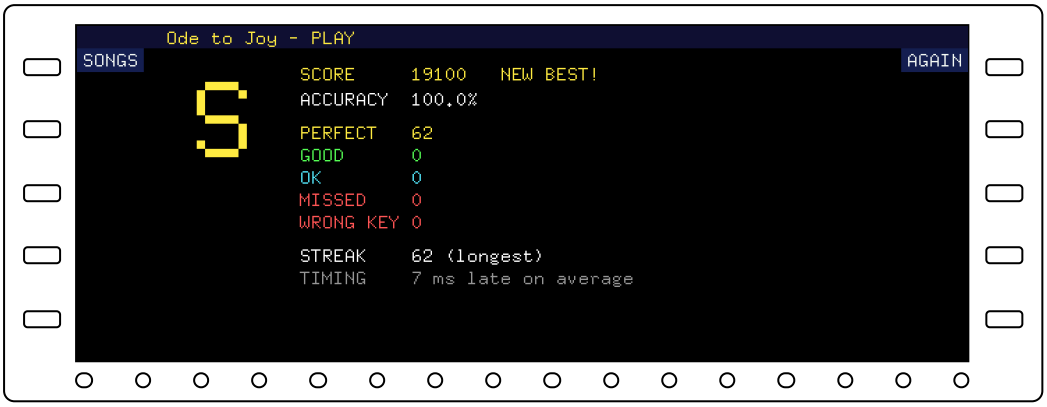
■ Stopping

Press **EXIT** to stop the song at once; the results of what was played follow. Holding **EXIT** leaves Keyfall altogether.

■ The accompaniment

The song's other parts go to the instrument's MIDI parts PART 4 to PART 16 (channel 10 the drums), as when a computer plays a song through **MIDI IN**. Their sounds are the General MIDI ones the song names. Keep the instrument's MIDI settings as they come (MIDI menu, **PART SETTING**: PART 4 to 16 on channels 4 to 16).

Results and best scores



The results of a game: the grade, the score, the counts, the longest streak and the average timing.

The grade comes from the accuracy: **S** from 95%, **A** from 85%, **B** from 70%, **C** from 50%, **D** below. **TIMING** says how early or late your keys came on average: a steady lateness you cannot hear is better set with **LATENCY** on the song list.

A game played to the end in PLAY mode that beats the best score for that song, part and tempo is saved in SCORES.TXT ("NEW BEST!"); the song list shows the best as **BEST**. WAIT shows how many notes were played and how many keys were wrong; DEMO, nothing to count.

LCDR 1 (AGAIN) plays the same song again; **LCDL 1** (SONGS) or **EXIT** returns to the song list.

Adding songs

Copy any Standard MIDI File (format 0 or 1) into the card's KEYFALL\ folder, with a short name ending in .MID (eight letters or digits at most before the dot). Each track's notes on one channel become a part named after the track. A track named "Right hand" (or RH, Melody) is offered first; with one named "Left hand" (or LH) as well, "Both hands" can be chosen.

System exclusive messages in a file are left out, so that a song cannot change the instrument's settings. Everything else of the other parts (notes, program changes, volume, pan, the sustain pedal, pitch bend) is sent.

The songs supplied: *Twinkle, Twinkle, Little Star* (traditional), *Ode to Joy* (Beethoven) and the *Minuet in G* (Petzold), each with its right hand and an accompaniment. They are written as text and turned into MIDI files by tools/songs/mksong.py, whose README describes the notation.

Troubleshooting

what happens	why, and what to do
Keys of the left side count for nothing.	**AUTO PLAY CHORD** is on: those keys choose chords and send no notes. Turn it off.
Every key comes late (TIMING in the results).	Raise **LATENCY** by about that much.
The accompaniment is silent or plays odd sounds.	The MIDI parts PART 4-16 are off or on other channels (MIDI menu, **PART SETTING**), or **LOCAL**/receive settings were changed. The initialized MIDI settings play it.
"This file cannot be played".	Not a MIDI file Keyfall reads (SMPTE timing, format 2), or no notes outside the drums.
A blue screen.	The game stopped: photograph the screen, switch the instrument off and on. KFLOG.TXT on the card tells what it did last.

What Keyfall changes in the instrument

Nothing is stored in the instrument: no panel memory, setting or file but SCORES.TXT and KFLOG.TXT in the game's own folder.

While the accompaniment plays, the MIDI parts PART 4 to PART 16 receive program changes, bank selects, volume, pan and expression, as from any sequencer. When the song ends, or Keyfall is left, every channel it used gets Reset All Controllers and All Notes Off; the sounds chosen stay on those MIDI parts until they receive others or the instrument is switched off. The keybed's own parts (RIGHT 1, RIGHT 2, LEFT) receive nothing.

The TEMPO/PROGRAM wheel and the soft keys beside the display are Keyfall's while it runs: the tempo does not change when the wheel chooses a song.